

Self-Certifying Return*

Eternal Return table

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Abstract

This note separates Poincaré recurrence from self-certifying readings of “this life returns.” In a measure-preserving system of finite measure, almost every point of a positive-measure set returns to that set infinitely often; when the space is separable and metric, almost every point returns to every neighborhood of itself (ε -Return). Exact identity of a continuous state is a different statement, and it is typically not a predicate of a recursively axiomatized arithmetic S : S can express equality of codes and coded neighborhood membership. Let $\Box$ be provability in a standard proof predicate for a consistent, recursively axiomatized extension S of Peano arithmetic. The logic of $\Box$ is Gödel–Löb logic GL. The scheme $\Box A \rightarrow A$ is not a theorem of GL; Löb’s theorem replaces it, and the Löb hypothesis is not the hypothesis of Poincaré’s theorem. A configuration that encodes S , or encodes a proof of its own return, is a self-referential certificate. The Henkin sentence for S is provable in S and does not prove Con_S . A fixed point that conjoins a recurrence formula with Con_S is unprovable if S is consistent. These are limits on self-certification inside S . They do not say that recurrent sets are empty. Separately, recurrence queries for configurations that encode a universal computation are undecidable. Undecidability of the query does not decide whether a given orbit returns. Nothing here equates ε -Return with the demand to will an identical life, and incompleteness of S is neither a refutation of the recurrence theorem nor an argument for it.

1 Introduction

Three different claims travel under the word “return.” This note gives each one a sentence, a system, and a hypothesis list.

The first is dynamical. A measure-preserving transformation of a finite measure space returns almost every point of a positive-measure set to that set, infinitely often [9, 13]. In a separable metric space the same argument yields *ε -Return*: almost sure return into every neighborhood of the point. The second is linguistic. A formal arithmetic S does not contain an arbitrary point of a continuum as a value of a variable. It contains codes. The predicates it can form of those codes are code-identity and coded neighborhood membership. The third is self-certifying. One asks the configuration to contain the system S , or a proof in S , and to count that included proof as establishing the return of the configuration that contains it. The tools for that reading are the second incompleteness theorem [3], Löb’s theorem [6], and the provability logic GL [1, 10]. Throughout, $\Box$ means provability in one named system S , not an unspecified necessity and not a knower.

*Packaging note for the Eternal Return table. The recurrence theorem is Poincaré’s, the incompleteness theorems are Gödel’s, the fixed-point theorem for provability is Löb’s, and the modal completeness theorem is Solovay’s. The note only keeps their quantifiers from being exchanged. It is the formal companion to the table review in the repository root.

The note proves no new incompleteness theorem and no new recurrence theorem. Section 2 records ε -Return with its hypotheses and the predicate boundary. Section 3 packages $\square$ as GL. Section 4 says why an S5 knowledge operator is a different symbol. Section 5 applies the second theorem and Löb's theorem to two named fixed points. Section 6 records a halting reduction for the query “does C recur?” Section 7 lists the readings that these results do not support. Gödel's rotating cosmological models [4] are a solution of Einstein's equations with closed timelike curves. They are not an argument about Prov_S , and they are not used below.

2 Setup: dynamics and the formal language

Two systems are in play, and they are not the same object. The dynamical statement lives in analysis. The provability statement lives in arithmetic.

Definition 1 (Foundations). ZFC is Zermelo–Fraenkel set theory with choice, used only as a sufficient ambient theory in which Lebesgue measure and the argument of Theorem 3 can be formalized. S is a consistent, recursively axiomatized extension of Peano arithmetic PA. The symbol $\square$ is never used for provability in ZFC. When provability in ZFC is intended it is written Prov_{ZFC} .

Consistency of S is an external assumption about S , in the usual sense that no S -derivation ends in $0 = 1$. It is not an axiom of S .

2.1 Measure-preserving dynamics

Definition 2. A *measure-preserving system* is a quadruple $(X, \mathcal{B}, \mu, T)$ where $(X, \mathcal{B}, \mu)$ is a measure space with $\mu(X) < \infty$ and $T: X \rightarrow X$ is measurable and satisfies $\mu(T^{-1}A) = \mu(A)$ for every $A \in \mathcal{B}$.

Finite measure is the hypothesis that does the work. Normalizing to a probability does not change the argument. In the classical physical reading, T is the time-1 map of a flow that preserves a finite measure on its region (Liouville measure on a finite-volume invariant set is the usual source of such a T). The note proves the discrete statement. A flow inherits integer-time returns from its time-1 map.

Theorem 3 (Poincaré recurrence). *Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system and let $A \in \mathcal{B}$ satisfy $\mu(A) > 0$. Then*

$$\mu(\{x \in A \mid \text{there are infinitely many } n \geq 1 \text{ with } T^n x \in A\}) = \mu(A).$$

Standard argument. Let $E = \{x \in A \mid \forall n \geq 1 (T^n x \notin A)\}$. The sets $T^{-k}E$ for $k \geq 0$ are pairwise disjoint. If $x \in T^{-i}E \cap T^{-j}E$ for $0 \leq i < j$, then $T^i x \in E \subseteq A$ and $T^{j-i}(T^i x) \in A$, contradicting $T^i x \in E$. Invariance gives $\mu(T^{-k}E) = \mu(E)$. Finite measure gives $\sum_{k \geq 0} \mu(E) \leq \mu(X) < \infty$, so $\mu(E) = 0$. Thus almost every point of A returns to A at least once.

The same argument applies to each iterate T^N , which is measure-preserving. The set of points of A that fail to return under every positive iterate of T^N has measure zero, and the set of points that never return at a time $n \geq N$ is contained in it. A countable union over N of null sets is null. $\square$

The theorem is a statement about almost every point of a named measurable set. It does not say that every point returns, and it does not say that the return time is short. Both strengthenings need hypotheses this argument does not have.

Definition 4 (ε -Return). Suppose in addition that X is a separable metric space, $\mathcal{B}$ contains the Borel σ -algebra, and d is the metric. The system has ε -Return when, for every $\varepsilon > 0$,

$$\mu(\{x \in X \mid \text{there are infinitely many } n \geq 1 \text{ with } d(T^n x, x) < \varepsilon\}) = \mu(X).$$

Corollary 5. *Every measure-preserving system on a separable metric space, with μ a finite Borel measure, has ε -Return.*

Proof. Let $\{U_i\}_{i \in \mathbb{N}}$ be a countable basis of nonempty open sets. By Theorem 3, for each i there is a null set $N_i \subseteq U_i$ of points that fail to return to U_i infinitely often. Let $N = \bigcup_i N_i$. If $x \notin N$ and $\varepsilon > 0$, separability supplies a basis element U_i with $x \in U_i \subseteq \{y \mid d(y, x) < \varepsilon\}$. Infinitely many returns of x to U_i are returns to that neighborhood. The null set N does not depend on ε . $\square$

ε -Return is neighborhood membership for almost every point. Exact identity $T^n x = x$ is a different predicate, and the finite-measure hypotheses do not force it.

Proposition 6 (Exact return can fail everywhere). *Let $X = \mathbb{R}/\mathbb{Z}$, let μ be Lebesgue measure, and let $T(x) = x + \alpha \bmod 1$ with α irrational. Then (X, μ, T) is a measure-preserving system, every point returns to every neighborhood of itself infinitely often, and $T^n x \neq x$ for every x and every $n \geq 1$.*

Proof. Translation preserves Lebesgue measure, and $\mu(X) = 1$. If $T^n x = x$, then $n\alpha \in \mathbb{Z}$, so α is rational. For the neighborhood claim, write $\|\cdot\|$ for distance to the nearest integer, so $d(T^n x, x) = \|n\alpha\|$. The $N + 1$ points $0, \{\alpha\}, \dots, \{N\alpha\}$ on the circle cannot be pairwise more than $1/N$ apart, so two of them lie at distance at most $1/N$. Their difference is an integer q with $1 \leq q \leq N$ and $\|q\alpha\| \leq 1/N$. Fix $\varepsilon > 0$ and suppose $F = \{n \geq 1 : \|n\alpha\| < \varepsilon\}$ is finite. Since α is irrational, $\|n\alpha\| > 0$ for every $n \geq 1$. If F is empty, any integer $N > 1/\varepsilon$ yields a q with $\|q\alpha\| < \varepsilon$, so $q \in F$, contradicting emptiness. If F is nonempty, let $\delta = \min_{n \in F} \|n\alpha\| > 0$ and choose an integer $N > \max(1/\varepsilon, 1/\delta)$. The same step yields $q \leq N$ with $\|q\alpha\| \leq 1/N < \delta$, so $q \in F$ and $\|q\alpha\| < \delta$, which contradicts the choice of δ . Thus F is infinite. Every point returns to every one of its neighborhoods infinitely often, and the system has ε -Return with empty exceptional set. This approximation argument is special to the example. The measure theorem that applies to every system in Definition 2 is Theorem 3, which is almost-everywhere. $\square$

Proposition 6 is the separator between the two dynamical predicates. The system meets every hypothesis of Corollary 5. The set of exactly periodic points is empty. A theorem that delivers ε -Return has not delivered bit-identity of states.

2.2 What S can say

Points of $\mathbb{R}^d$ are not natural numbers. The variables of S range over numbers, codes of formulas, and codes of proofs. A continuous state enters a sentence of S only after a code has been chosen: a Gödel number of a definition, an index of a computable Cauchy name, or a code of a basic open set.

Definition 7 (Code predicates). Fix a standard primitive recursive Gödel numbering. For codes c, c' and a positive rational q given by a code, the following are formulas of arithmetic:

- $\text{EqCode}(c, c')$, the atomic equality $c = c'$ of the codes as numbers;
- $\text{Near}(c, c', q)$, a formalized statement that the basic open sets (or Cauchy names) numbered by c and c' meet within tolerance q .

Neighborhood membership, once the neighborhood is coded, is of the first kind of statement. Identity of the underlying real, when both objects are presented by names, is agreement of those names, which is again a statement about the codes. Identity of binary expansions is not well-defined on reals: dyadic rationals have two expansions.

Remark 8 (Predicate boundary). Exact bit-identity of an uncoded continuous state is not an S -predicate. There is no formula $\varphi(v)$ of the language of S whose intended extension is a singleton $\{x\}$ for an arbitrary $x \in \mathbb{R}^d$, because there is no variable in that language for x . What S states, after a coding has been fixed, is EqCode or Near. Theorem 3, formalized in ZFC, is a sentence about measures and preimages. It is not a scheme of S -atoms “ x itself returns,” one for each real x .

A symbolic configuration that is already a natural number, or a finite word, is a code. Equality of two such configurations is EqCode, and S can speak about it directly. That case is the setting of Section 6. It does not restore bit-identity as an S -predicate of nameless reals, and it does not turn Theorem 3 into a self-certifying sentence.

Definition 9 (Proof predicate). Let $\text{Proof}_S(p, \ulcorner \varphi \urcorner)$ be a primitive recursive proof relation coming from a primitive recursive enumeration of the axioms of S , and set

$$\text{Prov}_S(\ulcorner \varphi \urcorner) :\iff \exists p \text{ Proof}_S(p, \ulcorner \varphi \urcorner), \quad \text{Con}_S :\iff \neg \text{Prov}_S(\ulcorner 0 = 1 \urcorner).$$

The box of this note is this predicate:

$$\Box \varphi :\iff \text{Prov}_S(\ulcorner \varphi \urcorner). \tag{1}$$

The Hilbert–Bernays–Löb derivability conditions are assumed for this choice [1, 5, 6]: for all sentences φ, ψ ,

$$\begin{aligned} \text{(D1)} \quad & \text{if } S \vdash \varphi \text{ then } S \vdash \text{Prov}_S(\ulcorner \varphi \urcorner), \\ \text{(D2)} \quad & S \vdash \text{Prov}_S(\ulcorner \varphi \rightarrow \psi \urcorner) \rightarrow (\text{Prov}_S(\ulcorner \varphi \urcorner) \rightarrow \text{Prov}_S(\ulcorner \psi \urcorner)), \\ \text{(D3)} \quad & S \vdash \text{Prov}_S(\ulcorner \varphi \urcorner) \rightarrow \text{Prov}_S(\ulcorner \text{Prov}_S(\ulcorner \varphi \urcorner) \urcorner). \end{aligned} \tag{2}$$

Condition (D1) is a rule about S , not an implication $\varphi \rightarrow \Box \varphi$ proved inside S . Conditions (D2) and (D3) are single theorems of S for each pair of sentences, in fact theorems of PA for the standard proof predicate.

Remark 10 (The predicate is intensional). The second incompleteness theorem applies to a proof predicate satisfying (2). It is not a theorem about every arithmetical formula that defines the same set of theorems in the standard model. There are numerations of the axioms of a consistent S for which the associated consistency statement is provable in S [2]. Any reading that changes the proof predicate has changed the claim. In this note Prov_S always means a standard predicate as in Definition 9.

3 Modal packaging

Gödel–Löb logic GL is the propositional modal logic of $\Box$ under (1) [1, 10]. Modal formulas are built from propositional variables by Boolean connectives and $\Box$. The axioms of GL are all propositional tautologies, the distribution axiom

$$\Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B),$$

and Löb's axiom

$$\Box(\Box A \rightarrow A) \rightarrow \Box A. \quad (3)$$

The rules are modus ponens and necessitation: from a theorem A , infer $\Box A$. Necessitation is a rule of proof. It does not add the theorem $A \rightarrow \Box A$. The transitivity scheme $\Box A \rightarrow \Box \Box A$ is derivable from (3); it is not an extra assumption.

An *arithmetical realization* is a map $*$ from propositional variables to sentences of S , extended by the Boolean clauses and by

$$(\Box A)^* = \text{Prov}_S(\ulcorner A^* \urcorner).$$

Theorem 11 (Solovay completeness). *Let S be a consistent recursively axiomatized extension of PA, and let Prov_S satisfy (2). A modal sentence A is a theorem of GL if and only if $S \vdash A^*$ for every arithmetical realization $*$.*

Citation. This is Solovay's arithmetical completeness theorem [10]; a full development is Boolos [1, Chapters 1–5]. The note uses it as a completed theorem. It does not reprove the Solovay embedding. $\square$

The scheme one would need in order to pass from “ S proves that it returns” to “it returns” is reflection, $\Box A \rightarrow A$. That scheme is not free in GL.

Proposition 12 (Reflection is not a theorem of GL). $\text{GL} \not\vdash \Box p \rightarrow p$. *On the Kripke frame consisting of a single world with no accessibility arrow, $\Box p \rightarrow p$ is not valid. Over S , the arithmetical instance $\text{Prov}_S(\ulcorner 0 = 1 \urcorner) \rightarrow 0 = 1$ is equivalent to Con_S .*

Proof. Finite transitive irreflexive frames validate GL [1]. In a one-point frame with empty accessibility relation, $\Box p$ is vacuously true at the only world, because that world sees no world at which p fails. Interpreting p as false at that world makes $\Box p \rightarrow p$ false there. Hence the scheme is not GL-valid, and not a GL-theorem.

For the arithmetical instance, PA proves $\neg(0 = 1)$. Propositional logic then yields equivalence, inside S , of $\neg \text{Prov}_S(\ulcorner 0 = 1 \urcorner)$ and $\text{Prov}_S(\ulcorner 0 = 1 \urcorner) \rightarrow 0 = 1$. The left-hand side is Con_S . $\square$

By Theorem 11 and the second incompleteness theorem (Theorem 16 below), S does not prove every arithmetical realization of $\Box p \rightarrow p$. The one-point frame shows the modal failure before any appeal to arithmetic. The two failures match.

Löb's axiom is the relation GL does prove. Read as a statement about a sentence R , it says:

$$\Box(\Box R \rightarrow R) \rightarrow \Box R. \quad (4)$$

The hypothesis is that the reflection instance for R is itself provable in S . The conclusion is that R is provable in S . Truth of R is a separate claim. Passing from $\Box R$ to R uses reflection, which Proposition 12 withholds.

There is a standard extension used for truth rather than for provability. GLS is GL together with the axiom scheme $\Box A \rightarrow A$, with necessitation restricted to theorems of GL [1]. The restriction is required for consistency of the calculus: unrestricted necessitation applied to the reflection instance $\Box \perp \rightarrow \perp$ yields $\Box(\Box \perp \rightarrow \perp)$, Löb's axiom yields $\Box \perp$, and reflection then yields $\perp$. Under a sound realization, theorems of GLS are true in the standard model. The reflection axioms of GLS are not theorems of S , and they are not theorems of GL. External soundness of S , when it is assumed, is an assumption about the standard model. It is not a derivation inside S , and it is not a hypothesis of Theorem 3.

Remark 13 (“Necessarily returns”). Take R to be an arithmetical sentence proposed as a coded return claim. The narrative inference “necessarily returns, therefore returns” is the inference from $\Box R$ to R . GL does not license it. The Löb condition under which S proves R is $S \vdash \text{Prov}_S(\ulcorner R \urcorner) \rightarrow R$, which by Löb’s theorem (Theorem 17) already yields $S \vdash R$. In that case the box is not an independent witness of return. S proves the return sentence outright, or the Löb hypothesis has not been met. The finite-measure hypotheses of Theorem 3 do not provide $S \vdash \Box R \rightarrow R$.

4 Epistemic packaging

The operator of this note is $\Box$, defined by (1) as provability in S . This section records what would change if the same box were reread as knowledge, and then declines that rereading.

The modal system **S5** for an operator K has distribution, the factivity axiom $KA \rightarrow A$, the positive-introspection axiom $KA \rightarrow KKA$, and the negative-introspection axiom $\neg KA \rightarrow K\neg KA$, with modus ponens and necessitation. Factivity is the inference “known return yields return.” Proposition 12 says that the corresponding scheme for Prov_S is not a theorem of GL and that its instance at $0 = 1$ is Con_S . A calculus that adds factivity for the same symbol that denotes Prov_S is the wrong calculus for that symbol. Calling the operator knowledge does not change the proof predicate.

A factive complex can be defined in the metatheory without pretending that S defines it:

$$K_S(\varphi) :\iff \text{Prov}_S(\ulcorner \varphi \urcorner) \text{ is true and } \varphi \text{ is true.} \quad (5)$$

The implication $K_S(\varphi) \rightarrow \varphi$ holds by the second conjunct, in the metatheory. It is not an S -theorem scheme. S cannot define a truth predicate for its own language [11], so (5) is not a formula of S . “Knowledge of return,” on this reading, is an external comment on a sentence. It is not a configuration in X , and it is not a proof of Con_S .

The note therefore does not adopt a knower, an **S5** axiom, or a knowability principle. Where a box appears below, it is Prov_S .

5 Self-certification barrier

Self-certification is a particular arithmetical sentence, obtained when a configuration code is required to contain S , or to contain a proof, and the sentence says of its own code that this happens. The diagonal lemma supplies the sentence. It does not by itself decide the sentence.

Lemma 14 (Diagonal lemma). *For every formula $\psi(v)$ of S with one free variable, there is a sentence ρ such that*

$$S \vdash \rho \leftrightarrow \psi(\ulcorner \rho \urcorner).$$

Citation. The usual fixed-point construction from the proof of the first incompleteness theorem [1, 3]. $\square$

Two fixed points have to be kept apart, and both have to be kept apart from ε -Return.

Definition 15 (Three sentences). Let $R(v)$ be any formula of S with one free variable. The intended reading of $R(c)$, when one is fixed, is an arithmetization of “the configuration coded by c recurs” or “the configuration coded by c encodes an S -proof of a named sentence.” The barrier statements do not depend on further details of R .

1. **External recurrence.** The sentence E of set theory is Theorem 3, including the finite-measure hypotheses in its antecedent. Its provability, if discussed, is $\text{Prov}_{\text{ZFC}}(\ulcorner E \urcorner)$.
2. **Henkin certificate.** By Lemma 14 applied to $\text{Prov}_S(v)$, there is a sentence σ with

$$S \vdash \sigma \leftrightarrow \text{Prov}_S(\ulcorner \sigma \urcorner). \quad (6)$$

This is the sentence that says “ S proves this sentence.”

3. **Consistency-bundled return.** By Lemma 14 applied to $R(v) \wedge \text{Con}_S$, there is a sentence ρ with

$$S \vdash \rho \leftrightarrow (R(\ulcorner \rho \urcorner) \wedge \text{Con}_S). \quad (7)$$

This is the sentence that says “the recurrence formula holds of my code, and S is consistent.”

Encoding S in a configuration is already covered by the second clause when the certificate is a proof code, and by the third when the sentence demands Con_S itself. A code for the axiom set of S can sit inside the number $\ulcorner \rho \urcorner$. That is ordinary Gödel numbering. It does not add an axiom to S .

Theorem 16 (Gödel’s second incompleteness theorem). *If S is consistent and Prov_S satisfies (2), then $S \not\vdash \text{Con}_S$.*

Citation. Gödel [3], in the form isolated by the derivability conditions [1, 5, 6]. The intensional caveat is Remark 10. □

Theorem 17 (Löb). *If $S \vdash \text{Prov}_S(\ulcorner \varphi \urcorner) \rightarrow \varphi$, then $S \vdash \varphi$. Equivalently, under (2),*

$$S \vdash \text{Prov}_S(\ulcorner \text{Prov}_S(\ulcorner \varphi \urcorner) \rightarrow \varphi \urcorner) \rightarrow \text{Prov}_S(\ulcorner \varphi \urcorner).$$

Citation. Löb [6]. The modal form is (3), via Theorem 11. □

Proposition 18 (The Henkin certificate does not prove consistency). *Assume S is consistent and Prov_S is standard. Then $S \vdash \sigma$, the sentence σ is true in the standard model, Con_S is true in the standard model, and*

$$S \not\vdash \sigma \rightarrow \text{Con}_S.$$

Proof. From (6), $S \vdash \text{Prov}_S(\ulcorner \sigma \urcorner) \rightarrow \sigma$. Theorem 17 gives $S \vdash \sigma$. A standard proof predicate represents actual provability, so $\text{Prov}_S(\ulcorner \sigma \urcorner)$ is true in $\mathbb{N}$, and therefore σ is true in $\mathbb{N}$. Consistency of S means that $0 = 1$ has no S -proof, so the standard sentence Con_S is true in $\mathbb{N}$. The implication $\sigma \rightarrow \text{Con}_S$ is therefore true in $\mathbb{N}$.

It is not a theorem of S . If $S \vdash \sigma \rightarrow \text{Con}_S$, then $S \vdash \sigma$ and modus ponens would give $S \vdash \text{Con}_S$, contradicting Theorem 16. □

Proposition 18 is the precise status of “this sentence is proved in S .” S does prove that self-certificate. Truth of the certificate in $\mathbb{N}$ is secured by the existence of the proof Löb’s theorem constructs. What fails inside S is the passage from that certificate to Con_S . The implication can be true in $\mathbb{N}$ and still unprovable in S . That is incompleteness of S at a true sentence. It is not a contradiction between the certificate and consistency, and it is not a statement about points of X .

Proposition 19 (The bundled fixed point is unprovable). *Assume S is consistent. Let ρ be as in (7) for an arbitrary formula $R(v)$. Then $S \vdash \rho \rightarrow \text{Con}_S$, and $S \not\vdash \rho$. If $R(v)$ is the formula $v = v$, then ρ is equivalent over S to Con_S , hence true in $\mathbb{N}$ and unprovable in S . If $R(v)$ is the formula $v \neq v$, then $S \vdash \neg \rho$. For a general recurrence formula, unprovability of ρ holds either way, and the truth-value of ρ in $\mathbb{N}$ tracks the truth-value of $R(\ulcorner \rho \urcorner)$.*

Proof. The left-to-right direction of (7) is $S \vdash \rho \rightarrow \text{Con}_S$. If $S \vdash \rho$, modus ponens yields $S \vdash \text{Con}_S$, contradicting Theorem 16. Hence $S \not\vdash \rho$.

If $S \vdash R(\ulcorner \rho \urcorner) \leftrightarrow \top$, then (7) gives $S \vdash \rho \leftrightarrow \text{Con}_S$. Consistency makes Con_S true and Theorem 16 makes it unprovable, so the same holds of ρ . If $S \vdash R(\ulcorner \rho \urcorner) \leftrightarrow \perp$, then $S \vdash \neg \rho$. In the general case, Con_S is true, so $\mathbb{N} \models \rho$ if and only if $\mathbb{N} \models R(\ulcorner \rho \urcorner)$. The note does not evaluate $R(\ulcorner \rho \urcorner)$ for any particular arithmetization of a dynamical return. $\square$

The slogan “this life returns, including this proof that it returns” collapses Definition 15. Split into the sentences actually available, it becomes one of the following, and they have different statuses.

Reading	Sentence	Status, when S is consistent
ε -Return	E in ZFC	Classical theorem under finite measure. Not a Con_S claim.
Henkin certificate	$\sigma \leftrightarrow \Box \sigma$	Provable in S . Does not prove Con_S .
Bundled return	$\rho \leftrightarrow R(\ulcorner \rho \urcorner) \wedge \text{Con}_S$	Unprovable in S . Truth tracks R , not a refutation of return.
Detachment	$\Box A \rightarrow A$	Not a theorem of GL. The instance $A = \perp$ is Con_S .

Remark 20 (No exemption by coding a configuration). Theorem 16 has no side condition of the form “unless a configuration code contains an index of S .” A numeral denoting such a code can appear in a sentence of S . If that sentence is Con_S itself, S does not prove it. If that sentence is σ , Proposition 18 applies. If that sentence is ρ , Proposition 19 applies. Writing a description of S into a number does not produce a derivation of Con_S from the axioms of S .

Remark 21 (Witnessed Σ_1 returns are a different sentence). A single coded orbit is not the bundled fixed point. If φ is a Σ_1 sentence and $\mathbb{N} \models \varphi$, then $\text{PA} \vdash \varphi$. In particular, if a computable configuration C actually returns to itself, a computation trace is a witness, and the Σ_1 sentence “ C recurs” is a theorem of PA. Incompleteness does not hide witnessed returns. The sentences that self-certification adds are of another kind: Con_S is Π_1 , reflection at a general sentence is not a true Σ_1 fact supplied by a return time, and ρ conjoins a recurrence formula with Con_S . Σ_1 -completeness settles the witnessed Σ_1 sentence and does not settle ρ .

Remark 22 (A stronger theory is not an internal certificate). Theorem 16 says that S does not prove Con_S . It does not say that no theory proves Con_S . A stronger system can prove the consistency of a weaker one. Whatever that stronger proof is, it is a derivation in the stronger system. It is not a derivation in S , and placing a copy of it inside a described configuration does not make it a derivation in S . The barrier in this section is self-certification: proving Con_S from within S , or detaching $\Box$ inside GL.

Nothing in Propositions 18 and 19 mentions μ , T , or a neighborhood. They remain true if the formula $R(v)$ arithmetizes a fact unrelated to dynamics. They therefore cannot cancel a measure calculation in ZFC, and they cannot supply the finite-measure hypothesis that Theorem 3 requires.

6 Undecidability of recurrence queries

The question “does configuration C recur?” is a decision problem about a named class of systems. Its undecidability, when it holds, is a fact about that decision problem.

Definition 23. A *Turing configuration* is a triple (q, t, h) where q is a state of a Turing machine, t is a tape inscription of finite support, and $h \in \mathbb{Z}$ is a head position. Step_M is the usual one-step function on configurations of a machine M . A configuration C *recurs* under M when there exists $n \geq 1$ with $\text{Step}_M^n(C) = C$.

Equality here is equality of finitely presented configurations. In the sense of Definition 7, it is EqCode, not neighborhood membership in a continuum.

Proposition 24 (Recurrence queries can encode halting). *The set of pairs (M, C) such that C recurs under M is not recursive.*

Standard reduction. The halting set $\{(N, w) \mid N \text{ halts on input } w\}$ is not recursive [12]. From a pair (N, w) , construct a machine $M_{N,w}$ whose finite control contains a copy of the transition table of N and a procedure that writes the finite word w . The map $(N, w) \mapsto M_{N,w}$ is computable. Let q_0 be the start state of $M_{N,w}$ and let C_0 be the configuration $(q_0, \text{blank tape}, 0)$. In the transition table, q_0 occurs only as the source of the first step and as the target of the cleanup step described below.

From C_0 the first transition enters a work state distinct from q_0 . The machine writes w , a separator, and a binary counter, then simulates N on w , incrementing the counter once per simulated step. Every configuration in this phase is distinct from C_0 : the state is not q_0 . If the simulation halts, the machine enters a cleanup state $q_{\text{clean}} \neq q_0$ and sweeps the finite marked region, erasing symbols until the tape is blank and the head is on cell 0, remaining in q_{clean} throughout the sweep. One further transition sends that blank-tape configuration to C_0 . A single Turing step is not asked to erase an arbitrary tape; the sweep does the erasing, and only the last step enters q_0 .

If N halts on w , the run from C_0 reaches that last step, so C_0 recurs. If N does not halt on w , the cleanup state is never entered, the counter is unbounded, and no later configuration has a blank tape in state q_0 . Thus C_0 does not recur. A decision procedure for recurrence of pairs (M, C) would decide the halting set by computing $(M_{N,w}, C_0)$. $\square$

The same style of encoding is what makes various long-term questions about dynamical systems that contain a universal computation undecidable [7]. Proposition 24 is the special case in which the long-term question is exact return of a finitely presented configuration. The short proof above is the direct reduction; it does not depend on the extra structure of generalized shifts.

Remark 25 (What the reduction produces). If N halts, the constructed C_0 recurs. If N does not halt, that C_0 does not recur. Both sorts of instance exist: there are machines that halt and machines that do not. Undecidability is the nonexistence of a total algorithm that sorts all instances. It is not an assertion that recurrent configurations do not exist, and it is not an assertion that they do.

Proposition 26 (Finitely presented systems are decidable). *Let X be a finite set and let $T: X \rightarrow X$ be given explicitly. For every $x \in X$ there exist $k \geq 0$ and $n \geq 1$ with $k + n \leq |X|$ and $T^{k+n}(x) = T^k(x)$. The point x itself recurs if and only if $T^n(x) = x$ for some n with $1 \leq n \leq |X|$. Both questions are decidable by running the orbit for at most $|X|$ steps.*

Proof. The $|X| + 1$ configurations $x, Tx, \dots, T^{|X|}x$ are not pairwise distinct, so $T^a(x) = T^b(x)$ for some $0 \leq a < b \leq |X|$. The first repetition closes a cycle. Running $|X|$ steps finds it. $\square$

Proposition 26 is the reason Proposition 24 needs an unbounded tape. The counter in the reduction takes infinitely many distinct values along a non-halting simulation. The counting measure on the set of all finite-support configurations is an infinite measure. Theorem 3 does not apply to that measure, so a non-recurrent C_0 is not a counterexample to ε -Return. A system whose

configuration set is finite falls under Proposition 26: every orbit eventually cycles, and the cycle is found by inspection. Exact eventual periodicity in a finite set is still EqCode on that set. It is not a proof of Con_S , and it is not ε -Return on a continuum.

7 What is not claimed

The positive content of the note is the separation recorded above. The following consequences do not follow from it.

1. **ε -Return stands under its own hypotheses.** Theorem 3 is a theorem of ordinary mathematics from finite measure and invariance. Its formalization is a theorem of ZFC under those hypotheses written into the antecedent. Consistency or incompleteness of S is not a hypothesis of the proof in Section 2, and it does not appear in the conclusion. Incompleteness of S does not entail that orbits fail to return.
2. **Incompleteness does not support the recurrence theorem.** Theorem 16 says $S \not\vdash \text{Con}_S$. It does not enlarge μ , it does not force $\mu(X) < \infty$, and it does not move a point into a neighborhood of itself. There is no inference from $S \not\vdash \text{Con}_S$ to ε -Return, nor from $S \not\vdash \text{Con}_S$ to exact identity of states. A failure of self-certification is not evidence that a phase point recurs.
3. **The bundled sentence is unprovable, and its truth tracks R .** Proposition 19 says that consistent S does not prove ρ . If the recurrence formula $R(\ulcorner \rho \urcorner)$ is true in $\mathbb{N}$, then ρ is true and unprovable. If it is false, then ρ is false. The note supplies no dynamical calculation that would decide $R(\ulcorner \rho \urcorner)$. Unprovability of the bundle is not a disproof of recurrence.
4. **Witnessed returns remain provable when they are Σ_1 and true.** Remark 21 is part of the claim. A trace of a return of a coded configuration is a proof in PA of that Σ_1 sentence. The self-certification barrier concerns certificates that ask S to prove Con_S , or ask GL to detach $\Box A \rightarrow A$, or ask a sentence to include Con_S in its own fixed point.
5. **$\Box$ is provability in S .** It is not an S5 knowledge operator and not a metaphysical necessity. “Necessarily returns, therefore returns” is $\Box R \rightarrow R$, which Proposition 12 excludes from GL. The admissible substitute is Löb’s condition $S \vdash \Box R \rightarrow R$, which yields $S \vdash R$ rather than a new passage from a box to a fact about X .
6. **ε -Return is not the demand to will an identical life.** The published ethical sentence asks whether one can will this life in the same order, again and innumerable times [8, §341]. ε -Return says that, under named measure hypotheses, almost every state returns to every neighborhood of itself. A neighborhood of a state is membership in a positive-radius set. Proposition 6 meets the measure hypotheses at every point and meets exact identity at none. Uniqueness of one biographical sequence is not a conclusion of Theorem 3. The ethical sentence is not a theorem of ZFC derived here, and it is not a theorem of GL.
7. **Gödel 1949 is a different construction.** The rotating universes of Gödel [4] are exact solutions of Einstein’s field equations in which some worldlines are closed. That is a geometric statement under those field equations and those metrics. It is not ε -Return, it is not Con_S , and it is not Proposition 24.

8. **Undecidability of a query is not a theorem about a measure.** Proposition 24 says that no algorithm classifies all machine-configuration pairs. Remark 25 records that the reduction builds recurrent configurations and non-recurrent ones. Proposition 26 records that a fully presented finite system is decidable by inspection and eventually periodic. None of these sentences is the negation of Theorem 3, and none of them is that theorem.

The three readings in the introduction can now be stated in one place, with the systems named. ε -Return is Theorem 3 and Corollary 5, proved in analysis from finite measure, and formalizable in ZFC. What S can say about a continuous state is EqCode or Near for a chosen code, as in Remark 8. What S can say about a certificate that includes S is Proposition 18 for the Henkin sentence and Proposition 19 for the sentence that conjoins Con_S ; the passage from $\Box R$ to R is Remark 13. The decision problem for exact recurrence of Turing configurations is Proposition 24. Each sentence keeps the hypotheses of its own proof. None of them is a booster for another.

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